Homework 2

for (CSED490V) Foundations of Continuous Computation

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Let $\mathbb{F}$ be the set of finite floating-point numbers with E-bit exponent and M-bit mantissa. Let $\varepsilon := 2^{-M-1}$ be the machine epsilon of $\mathbb{F}$, and $e_{\min} := -2^{E-1} + 2$ be the minimum exponent of $\mathbb{F}$. Let $\overline{\mathbb{F}} := \mathbb{F} \cup \{-\infty, +\infty, \mathrm{NaN}\}$. Throughout the problems, assume $E, M \geq 3$.

Problem 1. This problem studies interesting properties of floating-point numbers and operations.

(a) Let $\gamma^{-1} \colon \mathbb{F} \to \{0,1\}^{1+E+M}$ be the inverse of $\gamma \colon \{0,1\}^{1+E+M} \to \overline{\mathbb{F}}$ (defined in class) on $\mathbb{F}$, where we use $\gamma^{-1}(0) := (0 \ldots 0)$. For $x \in \mathbb{F}$, let

$$\mathrm{succ}(x) := \min\{y \in \mathbb{F} \cup \{+\infty\} \mid y > x\}.$$

Show that for all $x \in \mathbb{F}$ with $x \geq 0$,

$$\gamma^{-1}(\mathrm{succ}(x)) = \gamma^{-1}(x) + 1,$$

where $+$ denotes addition of bit strings as binary numbers. You must consider every case of x.

Remark. This result shows that the successor of a nonnegative float can be obtained by a simple bit addition operation.

(b) Find $a, b, c \in \mathbb{F}$ such that $(a \otimes b) \otimes c \neq a \otimes (b \otimes c)$. You should write a, b, c as functions of M.

(c) Show that for all $a \in [1,2) \cap \mathbb{F}$, $1 \oslash a$ is "almost" a multiplicative inverse of a:

$$a \otimes (1 \oslash a) \in \{1, 1 - \varepsilon\}.$$

Hint. Analyze the range of $a(1 \oslash a)$.

Remark. There exists a float in $[1,2)$ that has no multiplicative inverse: $a \otimes b \neq 1$ for some $a \in [1,2) \cap \mathbb{F}$ and all $b \in \mathbb{F}$. This highlights one difference from real arithmetic, where every real number in $[1,2)$ does have a (unique) multiplicative inverse.

(d) Let $\mathrm{sqrt}(x) := \mathrm{rnd}(\sqrt{x})$ for $x \in \mathbb{F}$ with $x \geq 0$. Show that for all $x \in \mathbb{F}$,

$$\mathrm{sqrt}(x \otimes x) = |x|.$$

Assume that no overflow and underflow occur in $x \otimes x$.

Solution. Throughout, let $e_{\max} = 2^{E-1} - 1$, $\Delta \coloneqq 2^{e_{\min}-M}$, and $F_{\max} = (2 - 2^{-M})2^{e_{\max}}$, as in the Course Notes. The rounding mode is round-to-nearest, ties-to-even.

(a) Nonnegative finite floats have sign bit zero, so their encodings are ordered lexicographically by exponent field and then by fraction field. We check the cases explicitly.

(1) $x = 0$: the convention gives the all-zero encoding, and the next bit string encodes Δ, the least positive subnormal.

(2) x positive subnormal with fraction field smaller than $2^M - 1$: incrementing that field gives the next multiple of Δ.

(3) $x = (2^M - 1)\Delta$: adding one gives exponent field 1 and zero fraction, which encodes the least positive normal $2^{e_{\min}} = 2^M\Delta$.

(4) x normal with fraction field smaller than $2^M - 1$: increasing the fraction field by one gives the next float in the same binade.

(5) x normal with maximal fraction field and nonmaximal finite exponent field: the carry increases the exponent field and sets the fraction to zero, producing the next binade's least float.

(6) $x = F_{\max}$: the carry changes the all-but-maximal exponent field to the all-ones exponent field with zero fraction. This is $+\infty = \operatorname{succ}(F_{\max})$.

Every transition is exactly one increment of the unsigned bit string; these cases exhaust $x \geq 0$. At $x = F_{\max}$, the displayed $\gamma^{-1}(+\infty)$ uses the evident extension of the inverse to the unique positive-infinity bit pattern (the source statement restricts its displayed domain to finite values).

(b) Set $h \coloneqq 2^{-M}$ and choose

$$a = b = 1 + h, \qquad c = \frac{3}{2}.$$

All three are representable when $M \geq 3$. Since $(1 + h)^2 = 1 + 2h + h^2$, and $0 < h^2 < h/2$, ties-to-even gives $a \otimes b = 1 + 2h$. Thus

$$(a \otimes b) \otimes c = \frac{3}{2} + 3h.$$

On the other hand, $bc = \frac{3}{2} + \frac{3}{2}h$ is the midpoint between $\frac{3}{2} + h$ and $\frac{3}{2} + 2h$; the even significand is the upper endpoint, so $b \otimes c = \frac{3}{2} + 2h$. Hence

$$a(b \otimes c) = \frac{3}{2} + \frac{7}{2}h + 2h^2.$$

This rounds to $\frac{3}{2} + 4h$: it lies strictly above the midpoint $\frac{3}{2} + \frac{7}{2}h$, and below the next midpoint because $2h^2 < h/2$. Therefore

$$(a \otimes b) \otimes c = \frac{3}{2} + 3h \neq \frac{3}{2} + 4h = a \otimes (b \otimes c).$$

(c) Write $u = 1/a$ and $b = 1 \oslash a = \operatorname{rnd}(u)$. Because $1/2 < u \leq 1$, the nearest-float error in this range is at most $2^{-M-2} = \varepsilon/2$. Consequently

$$|ab - 1| = a|b - u| \leq a\varepsilon/2 < \varepsilon, \qquad 1 - \varepsilon < ab < 1 + \varepsilon.$$

The float immediately below 1 is $1 - \varepsilon$, while the next float above 1 is $1 + 2\varepsilon$. Every real in $(1 - \varepsilon, 1 + \varepsilon)$ rounds to either $1 - \varepsilon$ or 1 (the midpoint between 1 and its successor is $1 + \varepsilon$). Thus $a \otimes (1 \oslash a) \in \{1 - \varepsilon, 1\}$.

(d) Put $u = |x|$. The case $u = 0$ is immediate. The no-underflow assumption forces u^2 into the normal range; since $e_{\min} < 0$, it also forces u to be normal. Write $2^e \leq u < 2^{e+1}$ and put

$h \coloneqq 2^{e-M}$, the spacing immediately above 2^e. If $u = 2^e$, then $u^2 = 2^{2e}$ is representable: the hypotheses ensure its exponent is in range. Hence $u \otimes u = u^2$ and the claim follows. Otherwise $u > 2^e$, and the adjacent floats on either side of u are $u - h$ and $u + h$ (if $u + h$ would overflow, the square already could not be finite). The corresponding midpoints in the square-root variable are $u - h/2$ and $u + h/2$. By the Course Notes' ULP bound, rounding u^2 produces $t = u \otimes u$ satisfying

$$|t - u^2| \le \tfrac{1}{2}\,\mathrm{ulp}(u^2) \le 2^{2e-M}.$$

Here the final inequality uses $u^2 < 2^{2e+2}$, including the binade boundary. Since $u \ge 2^e + h$,

$$u^2 - (u - h/2)^2 = uh - h^2/4 > 2^{2e-M},$$
$$(u + h/2)^2 - u^2 = uh + h^2/4 > 2^{2e-M}.$$

It follows that $(u - h/2)^2 < t < (u + h/2)^2$, hence $u - h/2 < \sqrt{t} < u + h/2$. Therefore $\mathrm{rnd}(\sqrt{t}) = u = |x|$, as required.

□

Problem 2. This problem studies two ways of computing the mean of $a, b \in \mathbb{F}$:

$$\widehat{m} \coloneqq (a \oplus b) \oslash 2, \qquad \widehat{m}' \coloneqq (a \oslash 2) \oplus (b \oslash 2).$$

(a) Show that if $|a + b| \le \max \mathbb{F}$ and $a \le b$, then

$$a \le \widehat{m} \le b.$$

(b) Show that if $|a + b| \le \max \mathbb{F}$, then $\widehat{m}$ is the correctly-rounded result:

$$\widehat{m} = \mathrm{rnd}\left(\frac{a+b}{2}\right).$$

Also compute the worst-case ULP error of $\widehat{m}$, and find a, b achieving this error.

Hint. Consider two cases: $|a + b| \ge 2^{e_{\min}+1}$ and otherwise.

(c) Show that if $|a| \ge 2^{e_{\min}+M+3}$, then $\widehat{m}'$ is the correctly-rounded result:

$$\widehat{m}' = \mathrm{rnd}\left(\frac{a+b}{2}\right).$$

Also prove that overflow cannot occur.

Remark. Part (a) shows that $\widehat{m} \in [\min(a, b), \max(a, b)]$, a desirable property for any mean computation. Parts (b) and (c) show that the best possible approximation of the mean can be obtained by evaluating $\widehat{m}$ when $|a|, |b|$ are both not too large (i.e., less than $2^{e_{\min}+M+3}$), and by using $\widehat{m}'$ otherwise.

Solution.

(a) By part (b), proved below, $\widehat{m} = \mathrm{rnd}((a + b)/2)$. Thus it is enough to check that rounding a number between two representable endpoints cannot pass an endpoint. The exact mean satisfies $a \le (a + b)/2 \le b$, and rnd is nondecreasing and fixes $a, b \in \mathbb{F}$ (Lecture 6, and the corresponding CN proposition). Hence

$$a = \mathrm{rnd}(a) \le \mathrm{rnd}\left(\frac{a+b}{2}\right) \le \mathrm{rnd}(b) = b.$$

(b) Put $t = a + b$ and $s = \mathrm{rnd}(t) = a \oplus b$. Let $\Delta = 2^{e_{\min}-M}$ be the least positive subnormal spacing. Both a and b are integer multiples of Δ, so t is too. If $|t| < 2^{e_{\min}+1}$, every such multiple is representable; therefore $s = t$ and $\widehat{m} = \mathrm{rnd}(t/2)$.
If $|t| \geq 2^{e_{\min}+1}$, then $s/2$ is exactly representable: shifting its significand right one binary exponent place stays at or above the normal/subnormal boundary (the case $|s| = 2^{e_{\min}+1}$ gives $s/2 = 2^{e_{\min}}$). Thus $\widehat{m} = s/2$. Halving preserves the ordering of neighboring normal floats, their nearest-point cells, and the parity of their significands. The nearest-float choice (including a tie) for t therefore halves to the nearest-float choice for $t/2$. Consequently $s/2 = \mathrm{rnd}(t/2)$ also in this case.
Write $m = (a + b)/2$. Since $\widehat{m} = \mathrm{rnd}(m)$ and $|m| \leq F_{\max}$, the Course Notes' rounding estimate gives

$$\mathrm{err}_{\mathrm{ulp}}(m, \widehat{m}) \leq \frac{1}{2}.$$

The bound is attained by $a = 0$ and $b = \Delta$: $m = \Delta/2$ is exactly halfway between 0 and Δ, and ties-to-even gives $\widehat{m} = 0$. Since $\mathrm{ulp}(\Delta/2) = \Delta$, the ULP error is exactly $1/2$.

(c) Set $H \coloneqq 2^{e_{\min}+M+3}$ and suppose $|a| \geq H$. In particular $a/2$ is a representable float. If $|b| \geq 2^{e_{\min}+1}$, then $b/2$ is also exactly representable, so

$$\widehat{m}' = \mathrm{rnd}(a/2 + b/2) = \mathrm{rnd}\left(\frac{a+b}{2}\right).$$

Otherwise $|b| < 2^{e_{\min}+1}$, whence $|b/2| < 2^{e_{\min}}$ and $|\,\mathrm{rnd}(b/2)| \leq 2^{e_{\min}}$ by rounding monotonicity. Since $|a/2| \geq 2^{e_{\min}+M+2}$, the distance from $a/2$ to the nearest rounding midpoint in either direction is at least $2^{e_{\min}}$ (it is exactly this at the lower-side midpoint of a binade boundary). It follows that both $a/2 + b/2$ and $a/2 + \mathrm{rnd}(b/2)$ round to $a/2$. Therefore $\widehat{m}' = \mathrm{rnd}((a + b)/2)$ again.
Finally, by monotonicity, for every $u \in \mathbb{F}$, $|\,\mathrm{rnd}(u/2)| \leq F_{\max}/2$; indeed $F_{\max}/2$ is representable and fixes itself under rounding. Hence the exact sum of the two halved operands has magnitude at most $F_{\max}$. No operation in $\widehat{m}'$ can overflow.

□

Problem 3. This problem studies the variance of $x_1, \ldots, x_n \in \mathbb{F}$:

$$s \coloneqq \frac{1}{n}\sum_{i=1}^{n}\left(x_i - \frac{1}{n}\sum_{j=1}^{n} x_j\right)^2.$$

Let $\widehat{s} \in \mathbb{F}$ be the result of evaluating s under floating-point arithmetic (i.e., $\oplus, \ominus, \otimes, \oslash$ are used for $+, -, \times, /$, and each sum is evaluated from left to right). Assume that no overflow and underflow occur in all floating-point operations.

(a) Show that

$$(x_1 \oplus x_2) \oplus \cdots \oplus x_n = \sum_{i=1}^{n} x_i(1 + \delta_i) + O(\varepsilon^2)$$

for some $\delta_1, \ldots, \delta_n$ with $|\delta_i| < (n-1)\varepsilon$. You may ignore all ε^2 terms as $O(\varepsilon^2)$.

(b) Show that

$$\frac{|s - \widehat{s}|}{|s|} \leq (n+3)\varepsilon + O(\varepsilon^2).$$

Problem 4. This problem studies an accurate method for computing the sum of floats, called the Kahan summation algorithm. It computes $x_1 + \cdots + x_n$ as follows:

```
def ksum(x1, ..., xn):
   (s, r) := (0, 0)
   for i = 1, ..., n:
      x'i := xi ⊕ r
      (s, r) := fast2sum(s, x'i)
   return s
```

where `fast2sum` is the algorithm from class. Assume that no overflow and underflow occur in all floating-point operations. Show that

$$\left|\mathrm{ksum}(x_1, \ldots, x_n) - \sum_{i=1}^{n} x_i\right| \le 3\varepsilon \sum_{i=1}^{n} |x_i| + O(\varepsilon^2).$$

You may ignore all ε^2 terms as $O(\varepsilon^2)$.

Remark. As seen in Problem 3(a), naive summation has a first-order error proportional to $(n-1)\varepsilon \sum_i |x_i|$, whereas the Kahan summation has first-order error $3\varepsilon \sum_i |x_i|$. Thus its first-order coefficient is independent of n.

Solution. We must account for the fact that the inputs to `fast2sum` need not be ordered by exponent. We first establish the estimate that for finite intermediate results,

$$(s, r) = \mathrm{fast2sum}(a, b) \quad \Longrightarrow \quad s + r = a + b + d, \qquad |d| \le \varepsilon|b| + O(\varepsilon^2)|b|. \tag{1}$$

If $e_a \ge e_b$, the Dekker theorem in Lecture 9 (and the CN) gives $d = 0$. Otherwise $|a| < |b|$, except that $a = 0$ is trivial. Write

$$s = \mathrm{rnd}(a + b), \quad t = \mathrm{rnd}(s - a), \quad r = \mathrm{rnd}(b - t).$$

The relative error estimate for addition and subtraction holds even in the subnormal range, since their exact results are integer multiples of the smallest subnormal spacing. Therefore

$$|s - (a + b)| \le \varepsilon|a + b| \le 2\varepsilon|b|, \qquad |s - a| \le (1 + 2\varepsilon)|b|.$$

It follows that

$$|t - b| \le |t - (s - a)| + |s - (a + b)| \le (3\varepsilon + 2\varepsilon^2)|b|.$$

For $M \ge 3$, the multiplier is less than $1/2$; thus t and b have the same sign and their magnitudes differ by a factor smaller than 2. Sterbenz's theorem makes $b \ominus t = b - t$ exact. Consequently

$$d = s + r - a - b = (s - a) - t, \qquad |d| \le \varepsilon(1 + 2\varepsilon)|b|,$$

which proves (1) without assuming a favorable operand order.

Let s_i, r_i be the values after iteration i, with $s_0 = r_0 = 0$, and let $b_i \coloneqq x_i \oplus r_{i-1}$. Define

$$\eta_i \coloneqq b_i - (x_i + r_{i-1}), \quad d_i \coloneqq (s_i + r_i) - (s_{i-1} + b_i), \quad \rho_i \coloneqq s_i - (s_{i-1} + b_i).$$

Applying the CN relative-error bound for addition and (1),

$$\begin{aligned} |\eta_i| &\le \varepsilon(|x_i| + |r_{i-1}|), \\ |d_i| &\le \varepsilon|b_i| + O(\varepsilon^2)|b_i|, \\ |\rho_i| &\le \varepsilon|s_{i-1} + b_i|. \end{aligned} \tag{2}$$

For each fixed n, induction over the loop gives

$$|s_i| \le \sum_{j=1}^{i} |x_j| + O(\varepsilon) \sum_{j=1}^{i} |x_j|, \qquad |r_i| = O(\varepsilon) \sum_{j=1}^{i} |x_j|. \tag{3}$$

Indeed $s_i = \mathrm{rnd}(s_{i-1} + b_i)$ and $|r_i| \le |\rho_i| + |d_i|$; (2) then gives the two bounds simultaneously by induction. In particular, (2)–(3) imply

$$|\eta_i| \le \varepsilon |x_i| + O(\varepsilon^2) \sum_{j=1}^{i-1} |x_j|, \qquad |d_i| \le \varepsilon |x_i| + O(\varepsilon^2) \sum_{j=1}^{i} |x_j|. \tag{4}$$

The exact telescoping identity

$$s_i + r_i = \sum_{j=1}^{i} x_j + \sum_{j=1}^{i} (\eta_j + d_j)$$

follows directly from the definitions. On the last iteration, instead of introducing d_n and subtracting r_n, use $s_n = s_{n-1} + b_n + \rho_n$. Thus

$$s_n - \sum_{j=1}^{n} x_j = \sum_{j=1}^{n-1} (\eta_j + d_j) + \eta_n + \rho_n.$$

By (3), $|\rho_n| \le \varepsilon \sum_{j=1}^{n} |x_j| + O(\varepsilon^2) \sum_{j=1}^{n} |x_j|$. Combining this estimate with (4) gives

$$\boxed{\left| s_n - \sum_{j=1}^{n} x_j \right| \le 3\varepsilon \sum_{j=1}^{n} |x_j| + O(\varepsilon^2) \sum_{j=1}^{n} |x_j|.}$$

As usual, the constants hidden in the second-order term may depend on the fixed length n. This proves the stated first-order bound.

□

Problem 5. This problem studies the computation of rounding errors arising from floating-point addition and multiplication. Assume that no overflow occurs in all the floating-point operations.

(a) Let $a, b \in \mathbb{F}$, and let $(s, r) := \mathrm{fast2sum}(a, b)$. Show that $e_a \ge e_b$ implies $s - a \in \mathbb{F}$, where e_x denotes the exponent of $x \in \mathbb{F}$ as in class.

(b) Find $a, b \in \mathbb{F}$ with $e_a < e_b$ such that $s + r \ne a + b$, where $(s, r) := \mathrm{fast2sum}(a, b)$. Write a, b as functions of M and $e_{\min}$.

(c) Find $a, b \in \mathbb{F}$ with $e_a + e_b = e_{\min} + M - 1$ such that $ab - (a \otimes b) \notin \mathbb{F}$. Write a, b as functions of M and $e_{\min}$.

Remark. Parts (b) and (c) show that the preconditions discussed in class for exactly computing rounding errors from $\oplus$ and $\otimes$ cannot be weakened.

Solution.

(a) Put $s := \mathrm{rnd}(a + b)$. By symmetry under simultaneous negation, we may assume $a \ge 0$. When $a = 0$, the hypothesis $e_a \ge e_b$ forces $e_b = e_{\min}$ under the zero-exponent convention of Lecture 9, and $s - a = b \in \mathbb{F}$ immediately. We henceforth take $a > 0$.
First suppose $b < 0$. If $e_a = e_b = e_{\min}$, then $|a|, |b| < 2^{e_{\min}+1}$, so $|a + b| < 2^{e_{\min}+1}$ and Hauser's exact-addition theorem implies $s = a + b$. If $e_a = e_b > e_{\min}$, both a and $-b$ are normal floats in the same binade and hence $a/2 < -b < 2a$. Sterbenz's theorem again gives

$s = a + b$. In either case $s - a = b \in \mathbb{F}$. If $e_a > e_b$, then $0 < -b < 2^{e_a} \le a$. When $-b \ge a/2$, Sterbenz makes $a + b$ exact. Otherwise, $e_a > e_{\min}$ makes $a/2 \in \mathbb{F}$ and $a/2 < a + b < a$. By monotonicity of rounding, $a/2 \le s \le a$; applying Sterbenz to a and s proves $s - a \in \mathbb{F}$.
Now suppose $b \ge 0$. If $e_a > e_b$, then $0 \le b < a$, so $a \le a + b < 2a$. When $2a \in \mathbb{F}$, monotonicity gives $a \le s \le 2a$; if $2a \notin \mathbb{F}$, then $e_a = e_{\max}$ and finite $s \le F_{\max} < 2a$. In both cases Sterbenz gives $s - a \in \mathbb{F}$.
Finally suppose $b \ge 0$ and $e_a = e_b = e$. Set $h \coloneqq 2^{e-M}$. The integer-significand lemma in the Course Notes shows that a, b are integer multiples of h and $0 \le b \le 2^{e+1} - h$, including when $e = e_{\min}$ and one or both operands are subnormal. The exact sum obeys $0 < a + b < 2^{e+2}$. Nearest rounding in this range has error at most h, and the rounded sum s is also an integer multiple of h. Consequently,

$$0 \le s - a \le b + |s - (a + b)| \le b + h \le 2^{e+1}, \qquad s - a \in h\mathbb{Z}.$$

Every nonnegative multiple of h no larger than 2^{e+1} is representable (by the integer-significand lemma, with 2^{e+1} itself allowed only when within the finite range). If $e = e_{\max}$, the same-sign case cannot yield a finite rounded sum; thus there is no exception at the largest exponent. We conclude $s - a \in \mathbb{F}$ in every case.

(b) Choose

$$a \coloneqq \Delta = 2^{e_{\min} - M}, \qquad b \coloneqq 1.$$

Then $e_a = e_{\min} < 0 = e_b$, and a is too small to affect the rounding of $a + b$ at 1: indeed $0 < a \le 2^{-M-2} < \frac{1}{2}\,\mathrm{ulp}(1)$ for $E, M \ge 3$. Consequently the algorithm computes

$$s = a \oplus b = 1, \qquad b' = s \ominus a = 1, \qquad r = b \ominus b' = 0.$$

Therefore $s + r = 1 \ne 1 + \Delta = a + b$.

□

Problem 6. This problem studies automatic analyses for bounding the range and absolute error of an expression. Consider an expression language similar to that discussed in class:

$$e ::= c \mid x_i \mid -e \mid e + e \mid e \times e \mid \sqrt{e},$$

where $c \in \mathbb{F}$ and $i \in \{0, \ldots, n-1\}$. Assume that only expressions that do not overflow and do not evaluate $\sqrt{\ }$ on negative inputs are considered. The floating-point semantics $[\![e]\!] \colon \mathbb{F}^n \to \mathbb{F}$ and real-valued semantics $[\![e]\!]_{\mathrm{r}} \colon \mathbb{F}^n \to \mathbb{R}$ are defined as in class. Design functions $\mathrm{range}(e, \Sigma) \subseteq \mathbb{R}$ and $\mathrm{error}(e, \Sigma) \in \mathbb{R}$ satisfying, for all $\sigma \in \Sigma$,

$$[\![e]\!](\sigma) \in \mathrm{range}(e, \Sigma), \qquad |[\![e]\!](\sigma) - [\![e]\!]_{\mathrm{r}}(\sigma)| \le \mathrm{error}(e, \Sigma).$$

For $\Sigma = [l_0, r_0] \times \cdots \times [l_{n-1}, r_{n-1}]$, the elementary cases are defined as follows:

$$\begin{aligned}
\mathrm{range}(c, \Sigma) &= [c, c], & \mathrm{error}(c, \Sigma) &= 0, \\
\mathrm{range}(x_i, \Sigma) &= [l_i, r_i], & \mathrm{error}(x_i, \Sigma) &= 0, \\
\mathrm{range}(-e_1, \Sigma) &= [-b, -a] & & \text{if } \mathrm{range}(e_1, \Sigma) = [a, b], \\
\mathrm{error}(-e_1, \Sigma) &= \mathrm{error}(e_1, \Sigma). & &
\end{aligned}$$

If $\mathrm{range}(e_i, \Sigma) = [a_i, b_i]$, the remaining given rules are

$$\begin{aligned}
\mathrm{range}(e_1 + e_2, \Sigma) &= [\mathrm{rnd}(a_1 + a_2), \mathrm{rnd}(b_1 + b_2)], \\
\mathrm{error}(e_1 + e_2, \Sigma) &= \mathrm{error}(e_1, \Sigma) + \mathrm{error}(e_2, \Sigma) \\
&\quad + \tfrac{1}{2}\,\mathrm{ulp}(\max\{|a_1 + a_2|, |b_1 + b_2|\})\,.
\end{aligned}$$

Assume additionally that all computations performed in range and error do not overflow or evaluate a real-valued or floating-point square root on a negative input. With this assumption, the displayed addition rule is simpler than the general version discussed in class.

(a) Define $\mathrm{range}(e_1 \times e_2, \Sigma)$ and $\mathrm{range}(\sqrt{e_1}, \Sigma)$, and prove their soundness.

(b) Define $\mathrm{error}(e_1 \times e_2, \Sigma)$ and $\mathrm{error}(\sqrt{e_1}, \Sigma)$, and prove their soundness. You may assume $[\![e]\!]_{\mathrm{r}}(\sigma) \in \mathrm{range}(e, \Sigma)$ for all e, Σ, σ, even though this need not hold in general.

Hint. Use $\sqrt{a} - \sqrt{b} = (a - b)/(\sqrt{a} + \sqrt{b})$ for $a, b > 0$.

(c) See `p6.ipynb`.

Solution. As in the latest CN, fix Σ and abbreviate $\mathrm{range}(e, \Sigma)$ to $\mathrm{range}(e)$. Similarly, write $\mathrm{err}(e)$ for the assignment's $\mathrm{error}(e, \Sigma)$. This is a change of notation only. All intervals below enclose floating-point semantics; part (b) uses the additional real-semantics hypothesis stated in the problem.

(a) Let $\mathrm{range}(e_i) = [a_i, b_i]$ and define

$$L \coloneqq \min\{a_1a_2, a_1b_2, b_1a_2, b_1b_2\},$$
$$U \coloneqq \max\{a_1a_2, a_1b_2, b_1a_2, b_1b_2\}.$$

Set

$$\boxed{\mathrm{range}(e_1 \times e_2) \coloneqq [\mathrm{rnd}(L), \mathrm{rnd}(U)].} \tag{1}$$

Indeed, if $u_i = [\![e_i]\!](\sigma)$, then $u_i \in [a_i, b_i]$. The ordinary real product of two numbers in these intervals belongs to $[L, U]$ (the extrema of a bilinear function on a rectangle occur at its corners). The CN monotonicity theorem therefore gives $\mathrm{rnd}(L) \le \mathrm{rnd}(u_1u_2) \le \mathrm{rnd}(U)$, as required.

For the square root, write $\mathrm{range}(e_1) = [a, b]$ and put $a_+ \coloneqq \max\{0, a\}$. The promised nonnegative square-root arguments ensure $b \ge 0$. Set

$$\boxed{\mathrm{range}(\sqrt{e_1}) \coloneqq [\mathrm{rnd}(\sqrt{a_+}), \mathrm{rnd}(\sqrt{b})].} \tag{2}$$

Every reachable $u = [\![e_1]\!](\sigma)$ satisfies $a_+ \le u \le b$. Both square root and nearest rounding are nondecreasing, proving soundness. Clipping the lower endpoint at zero is necessary when a conservative input interval extends below zero even though no reachable evaluation does.

(b) Put $E_i \coloneqq \mathrm{err}(e_i)$ and $U_i \coloneqq \max\{|a_i|, |b_i|\}$. For multiplication, define

$$\boxed{\mathrm{err}(e_1 \times e_2) \coloneqq U_1E_2 + U_2E_1 + E_1E_2 + \tfrac{1}{2}\,\mathrm{ulp}(U_1U_2).} \tag{3}$$

To justify it, write $u_i = [\![e_i]\!](\sigma)$ and $v_i = [\![e_i]\!]_{\mathrm{r}}(\sigma)$. Then $|u_i| \le U_i$ and $|u_i - v_i| \le E_i$, so $|v_i| \le U_i + E_i$. Therefore

$$\begin{aligned} |u_1u_2 - v_1v_2| &\le |u_1||u_2 - v_2| + |u_1 - v_1||v_2| \\ &\le U_1E_2 + E_1(U_2 + E_2). \end{aligned}$$

The real product has $|u_1u_2| \le U_1U_2$; by the CN rounding bound, $|\,\mathrm{rnd}(u_1u_2) - u_1u_2| \le \frac{1}{2}\,\mathrm{ulp}(U_1U_2)$. The triangle inequality proves (3).

For the square root, let $\mathrm{range}(e_1) = [a, b]$ and $b \ge 0$. One sound definition is

$$\boxed{\mathrm{err}(\sqrt{e_1}) \coloneqq \sqrt{E_1} + \tfrac{1}{2}\,\mathrm{ulp}(\sqrt{b}).} \tag{4}$$

Indeed, $u = [\![e_1]\!](\sigma) \in [0, b]$, and the additional hypothesis ensures $v = [\![e_1]\!]_{\mathrm{r}}(\sigma) \ge 0$. For $u, v \ge 0$,

$$|\sqrt{u} - \sqrt{v}| = \frac{|u - v|}{\sqrt{u} + \sqrt{v}} \le \sqrt{|u - v|} \le \sqrt{E_1};$$

the inequality also holds when one of u, v is zero, by direct inspection. The rounding of $\sqrt{u}$ contributes at most $\frac{1}{2}\,\mathrm{ulp}(\sqrt{u}) \le \frac{1}{2}\,\mathrm{ulp}(\sqrt{b})$, which proves (4). Both (3) and (4) are conservative rather than necessarily sharp.

(c) The provided `p6.ipynb` implements (1)–(4) by recursion over the expression tree, in addition to the given constant, variable, negation, and addition cases. The binary64 range functions use actual floating-point operations for endpoints; the error function converts those endpoints exactly to rationals and forms all bounds with rational arithmetic. An exact integer-square-root calculation produces a *rational upper enclosure* for the square roots in (4), so an inexact call to `sqrt_rational()` cannot invalidate the bound. The same method bounds the ULP at the square-root upper endpoint.
For the degree-n polynomial

$$p_n(x) = \sum_{j=0}^{n} c_j x^j, \qquad c_j = \mathrm{rnd}(1/j!), \qquad n \in \{2, 3, 4, 5\},$$

the notebook constructs Horner's nested evaluation and the separate power-sum expression, and runs the analysis on $x \in [-1, 1]$. The resulting absolute-error bounds, displayed here in decimal form for comparison, are

n	Horner	Power sum
2	$4.9960 \cdot 10^{-16}$	$6.6613 \cdot 10^{-16}$
3	$5.6899 \cdot 10^{-16}$	$9.3906 \cdot 10^{-16}$
4	$5.8634 \cdot 10^{-16}$	$1.1785 \cdot 10^{-15}$
5	$5.9067 \cdot 10^{-16}$	$1.4051 \cdot 10^{-15}$

The certified values are the exact `Rational` objects returned by the notebook, not the displayed decimal approximations. Horner's method has a smaller upper bound in each case because it requires fewer separately rounded products and handles accumulated terms within one nested expression. The separate power construction adds rounding errors and independently propagated interval widths.

□